Reevaluating Pitcher Aging Curves in the Statcast Era

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Introduction

In recent years, there has been a dramatic evolution of the pitching landscape within Major League Baseball. This evolution is strongly driven by advancements in tracking technology that has in turn developed into changes in training and a notable increase in pitch velocity. Since the introduction of Statcast in 2015, teams have been able to monitor and analyze pitching performance in unprecedented detail for some time, such as capturing metrics like velocity, spin rate, release point, and pitch movement. The ability to analyze and leverage this data now provides insights into pitcher performance, workload, and fatigue that were largely unavailable in years before the rise and era of modern technology. Alongside these technological advancements, pitchers are throwing harder than ever before and modern training regimens emphasize both strength and power. Thus, increasing the physical intensity of pitching at all levels of professional baseball. This evolution has coincided with rising injury rates, shorter careers for some pitchers, and a shift in how teams manage rotations and bullpen usage. This raises important questions about the typical age at which pitchers reach their peak performance.

Understanding whether pitchers now reach peak productivity at younger ages and experience faster decline is critical for player development, injury prevention, and long-term contract valuation. If modern pitchers peak earlier due to increased velocity and physical strain, organizations may need to adjust scouting, draft strategies, and minor league usage to maximize performance and career longevity. Recent studies have begun to explore how increased pitch velocity, higher physical demands, and advanced tracking metrics affect pitcher performance and career trajectories. Findings suggest that modern pitchers may be experiencing shifts in peak performance age and patterns of decline, although the results are not yet conclusive. The following review synthesizes these studies, organizing the literature around key themes such as aging curves, velocity trends, injury risk, and workload management, helping provide a clearer picture of how the Statcast era may have influenced pitcher development and longevity.

Literature Review

We can see that a recent study examines the individual factors associated with baseball pitching performance, focusing on ball velocity, accuracy, and spin characteristics using a scoping review of 34 cross sectional studies across multiple countries *(Mercier, Tremblay, Daneau, & Descarreaux, 2020)*. This study includes a wide variety of pitchers ranging fromMLB, college, high school, and youth. The review discovered that pitching performance is strongly influenced by kinematic and kinetic factors. Kinematic variables describe the movement patterns and body positioning that contribute to efficient energy transfer and ball velocity. This includes shoulder horizontal adduction, forward and lateral trunk tilt, and lead knee extension. On the other hand, kinetic factors are the magnitude of force generated during the pitching motion and are closely associated with both increased velocity and heightened stress on the throwing arm. This can be seen in proximal forces acting at the shoulder and elbow. There are also other physical characteristics that correlate with higher velocity like body weight, age, grip strength, sprinting ability, and jump tests. With youth pitchers typically producing lower velocity, yet improving rapidly as they mature physically.

This suggests that physical development plays a crucial role early in a pitcher's career, while technical skills are increasingly important at higher levels of play. Although the review does not examine changes in peak pitching age, the association between increased velocity, greater joint forces, and heightened injury risk supports the possibility that modern training intensity and performance demands may contribute to higher injury risk. This elevated injury risk has the potential to influence career longevity and the timing of peak performance, providing important context for how injuries will be considered when examining shifts in pitcher aging curves. However, the use of primarily cross-sectional designs limits inference, reinforcing the need for longitudinal analyses of pitcher aging in the Statcast era.

While the scoping review highlights how biomechanics and physical development influence pitching performance across levels, other research has more directly examined how aging affects pitching mechanics at the professional level. Using a 3D motion analysis system, another study examined the effects of age on pitching kinematics in professional baseball pitchers to better understand how biological aging influences throwing mechanics *(Dun, Fleisig, Loftice, Kingsley, & Andrews, 2006)*. The authors compared younger (<20.4 years) and older (>27 years) professional pitchers across 18 kinematic variables measured during key phases of the pitching motion. There were six significant position- based variables that differed between the groups. These variables are stride length, pelvis and upper trunk orientation at lead foot contact, maximum shoulder external rotation during the arm-cocking phase, and lead knee flexion and forward trunk tilt at ball release. In older pitchers, the study saw shorter stride lengths, more closed pelvis and trunk positions, reduced shoulder external rotation, increased lead knee flexion, and less forward trunk tilt, suggesting declines in flexibility and joint range of motion with age. Despite these mechanical differences, there were no significant differences found in ball velocity or segment angular velocities between age groups. This indicates that older professional pitchers may adapt their mechanics or coordination strategies to compensate for age-related biomechanical changes and maintain performance. Overall, these findings highlight that while aging alters pitching kinematics, performance outcomes such as velocity may be preserved through technical adaptation, underscoring the complex relationship between aging, mechanics, and sustained performance in professional pitchers. However, because this study compares age groups cross-sectionally and predates the Statcast era's emphasis on velocity maximization, it does not address whether the age at which pitchers reach peak performance has shifted earlier under modern training loads and physical demands. Therefore, longitudinal analyses using Statcast data are needed to see whether higher velocity and intensity have changed traditional pitching aging curves.

On the other hand, more relevant to my research, there have been longitudinal studies of Major League Baseball players that provide a baseline for understanding typical peak performance ages before the Statcast era. Using data spanning 86 seasons (1921–2006), researchers examined how pitchers and hitters age across different skills while accounting for rule changes, ballpark effects, and league-wide performance trends *("Peak Athletic Performance and Ageing: Evidence from Baseball," 2026)*. The findings indicate that pitchers generally reach peak performance around age 29, slightly later than some earlier estimates. Skills that rely more heavily on athleticism, such as running speed, hitting for average, and strikeout prevention, tend to peak earlier, whereas skills that depend on experience and strategy, such as drawing walks or controlling walks, peak later. Notably, Hall-of-Fame players often reach peak performance at slightly older ages and maintain it longer, demonstrating more gradual declines than their peers. These patterns illustrate the interaction between physical decline and experience-based compensation, showing that players can offset losses in athletic ability with skill and knowledge. Understanding these historical aging curves is directly relevant to my study because they provide a benchmark against which the impact of modern Statcast-era factors can be assessed. By comparing historical peak ages to those of contemporary pitchers, this research aims to determine whether the typical pitching peak has shifted earlier in response to the demands of modern professional baseball.

Building on historical analyses of aging in Major League Baseball, a much older study examined the relationship between age and performance in 388 players active in 1965, including 153 pitchers and 235 hitters, all with at least 10-year careers *("APA PsycNet," 2026)*. By focusing on longitudinal performance metrics such as ERA, strikeouts, wins, batting average, and fielding statistics, the authors identified peak ages for different skills and explored the factors shaping career trajectories. Overall, pitchers and hitters generally reached peak performance between ages 27 and 30, concurrent with the previous study mentioned. More specifically, skills dependent on athleticism, like strikeouts or stolen bases, peaked earlier, while skills relying on experience and strategy, like pitching wins, walks, or fielding, peaked later. These findings reinforce the notion that peak performance results from a combination of physical ability and experience, with decline in the 30s influenced by both aging and injury. Longitudinal analyses confirmed that experience accelerated early career improvement but could not fully compensate for physiological limits, highlighting the interplay of physical capacity, experience, and motivation in shaping performance. Overall, MLB performance follows an inverted backward-J curve, reflecting patterns of peak and decline seen in other skill-based domains.

Methodology

This study analyzes Major League Baseball pitcher aging patterns using both modern Statcast-era data and historical pitching data. The primary modern dataset consists of pitcher-season observations beginning in 2015 and includes variables such as ERA, age, innings pitched, pitch count, fastball velocity, spin rate, strikeout rate, walk rate, whiff rate, and role indicators such as starting share. The dataset was imported into R and cleaned using the tidyverse and janitor packages to standardize variable names and ensure consistent formatting. Unique identifiers were created for each pitcher, and additional variables were constructed to capture pitcher roles and usage patterns. For example, a *start_share* variable was calculated to measure the proportion of games in which a pitcher appeared as a starter relative to total appearances. To ensure that the analysis focused on meaningful pitcher seasons, observations were filtered to include only pitcher-seasons with at least 50 innings pitched, a minimum pitch sample size of 200 pitches, and valid fastball velocity data.

After cleaning the dataset, several standardized variables were constructed to represent pitching intensity and workload. Continuous performance variables such as fastball velocity, spin rate, pitch count, innings pitched, whiff rate, strikeout rate, and walk rate were standardized using z-scores. Standardizing these variables allows the statistical models to compare effects across variables measured on different scales and helps stabilize coefficient estimates within the regression models. These standardized variables were used as controls in the statistical models to isolate the effect of age on pitching performance while accounting for differences in pitch quality, workload, and pitcher usage.

To describe basic aging patterns before estimating statistical models, the study first conducts a raw peak-season analysis. For each pitcher in the dataset, the season with the lowest ERA was identified as that pitcher's peak performance season. If multiple seasons had the same ERA, ties were broken using innings pitched and total pitch count to prioritize seasons with larger workloads. Summary statistics and frequency tables were then calculated to examine the distribution of peak ages across pitchers. A histogram of peak ages was generated to visualize where peak performance tends to occur in a pitcher's career.

The first mixed-effects model estimates the baseline Statcast-era aging curve. In this model, ERA serves as the dependent variable, while age splines capture the nonlinear effect of age on performance. Additional explanatory variables are included to control for pitching characteristics and workload, including standardized fastball velocity, spin rate, pitch volume, innings pitched, whiff rate, and starting role share. Year fixed effects are also included to account for league-wide changes in run environments or pitching conditions across seasons. After estimating the model, predicted ERA values were generated for ages 21 through 40 while holding all control variables constant at their average values. This prediction grid was used to construct the model-based Statcast-era aging curve.

A second model extends the baseline specification by introducing an interaction between age and fastball velocity. This interaction tests whether pitchers with higher velocity experience different aging trajectories compared with pitchers who throw slower fastballs. Predicted aging curves were then generated for pitchers with low, average, and high velocity levels (defined as −1, 0, and +1 standard deviations from the mean). Comparing these predicted curves allows the analysis to determine whether velocity influences both peak performance and the rate of aging.

In addition to modeling average aging curves, the study also examines performance volatility across pitcher careers. To measure volatility, the year-to-year change in ERA was calculated for each pitcher by comparing ERA in consecutive seasons. The absolute value of this change was used as the dependent variable to capture the magnitude of performance fluctuations regardless of direction. A mixed-effects model similar to the aging curve model was then estimated using absolute ERA change as the dependent variable. This approach allows the analysis to determine whether pitcher performance becomes more unstable with age while controlling for workload and pitch characteristics.

To compare modern pitcher development with earlier baseball eras, a historical dataset was constructed using the Lahman Baseball Database. Pitching statistics from 1985 through 2000 were selected to represent a pre-Statcast environment. The Lahman pitching dataset was merged with player demographic information to calculate pitcher ages. Key variables including ERA, innings pitched, strikeout rate, and walk rate were constructed to match the performance variables used in the modern dataset as closely as possible. A mixed-effects aging model similar to the Statcast-era model was then estimated using the historical data, allowing the study to estimate a pre-Statcast aging curve.

Additional historical analysis was conducted by dividing pitchers into strikeout ability tiers based on their strikeout rates. Pitchers were grouped into low, medium, and high strikeout categories using quantile-based classification. A mixed-effects model with an interaction between age and strikeout tier was then estimated to determine whether high-strikeout pitchers exhibit different aging patterns compared with lower-strikeout pitchers.

Finally, a combined comparison model was constructed to directly compare pitcher aging patterns across eras. To ensure comparability between datasets, only variables available in both the Statcast dataset and the Lahman dataset were included in this model. These variables include ERA, strikeout rate, walk rate, innings pitched, and pitcher age. The combined dataset includes both historical and modern observations and includes an indicator variable representing the era (Statcast vs. pre-Statcast). The mixed-effects model then interacts the age spline terms with the era indicator, allowing the estimated aging curve to differ between eras. Predicted aging curves were generated from this model to visually compare performance trajectories across the two baseball environments and to estimate peak performance ages and post-peak decline rates in each era.

Results and Discussion

To refresh, the primary goal of this study is to estimate pitcher aging curves while controlling for modern performance variables such as intensity, workload, pitch characteristics, and strikeout ability. Using mixed-effects models, the analysis isolates the relationship between age and pitcher performance while accounting for differences across individual pitchers and seasons. The Statcast-era model estimates the relationship between age and ERA while holding other performance variables constant, including fastball velocity, spin rate, pitch volume, innings pitched, and whiff rate. The resulting aging curve, as shown in *Figure 1,* suggests that pitcher performance improves slightly early in a career before gradually declining later in life. When workload and intensity variables are held constant, predicted ERA remains relatively stable through the mid-to-late twenties before beginning a gradual upward trend as pitchers age. This pattern suggests that physical decline and performance deterioration occur slowly and steadily rather than abruptly.

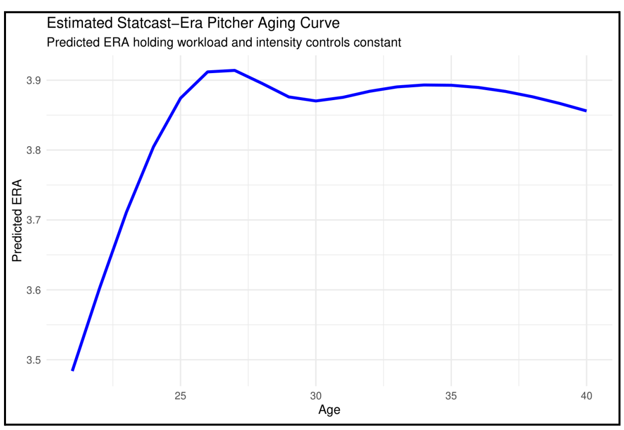

Figure 1: Estimated Statcase-Era Pitcher Aging Curve

The Statcast-era aging curve shows a relatively flat performance trajectory during a pitcher's prime years, with predicted ERA values remaining close together across the late twenties and early thirties. This indicates that once pitchers reach peak physical maturity, their performance can remain relatively stable for several seasons when other factors such as workload and pitch quality remain constant. However, the curve begins to rise in later years, indicating worsening run prevention as pitchers move into their mid-to-late thirties.

To further examine the role of pitching characteristics in aging patterns, a second mixed-effects model incorporates an interaction between age and fastball velocity as seen below in *Figure 2*. This model allows the aging trajectory to differ across pitchers with varying velocity levels. The results show that pitchers with higher fastball velocity maintain consistently lower predicted ERAs across most of the age range compared with pitchers with lower velocity. Although the statistical test comparing the interaction model to the baseline model suggests that the interaction is not strongly significant, the graphical patterns still indicate that velocity provides a persistent advantage in run prevention. High-velocity pitchers begin their careers with lower ERAs and maintain stronger performance as they age.

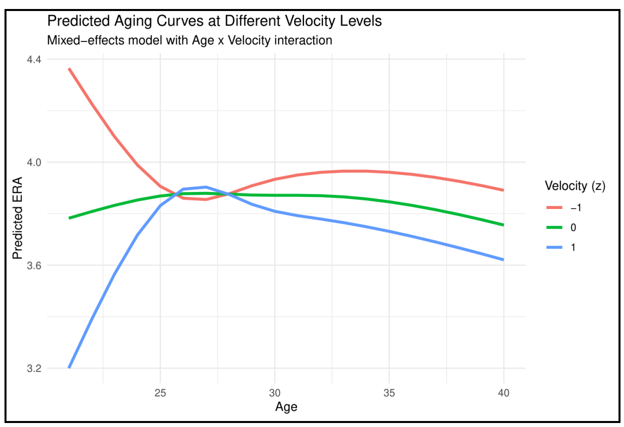

Figure 2: Predicted Aging Curves at Different Velocity Levels

The velocity-based aging curves illustrate that pitchers with below-average velocity experience higher predicted ERA levels throughout their careers, while pitchers with above-average velocity maintain better performance. This finding supports the widely held view that velocity is one of the most important indicators of pitcher effectiveness. While all pitchers experience performance decline with age, the decline appears less severe for pitchers with stronger velocity profiles.

Another dimension of pitcher aging examined in this study is performance volatility, defined as the year-to-year absolute change in ERA. Volatility provides insight into the stability of pitcher performance across seasons. The volatility model estimates the relationship between age and performance variability while controlling for workload, pitch characteristics, and role variables.

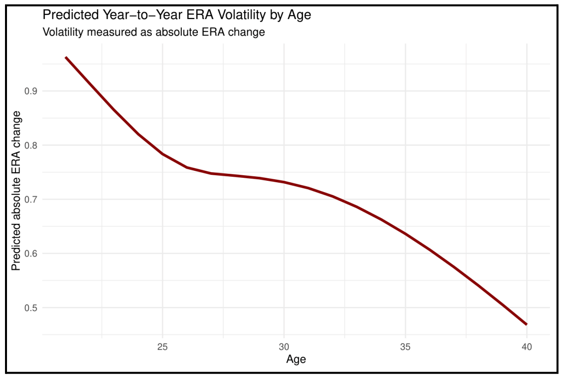

Figure 3: Predicted Year-to-Year ERA Volatility by Age

As seen in *Figure 3,* the results indicate that year-to-year ERA variability increases modestly with age. Younger pitchers tend to have more stable performance trajectories, while older pitchers exhibit greater fluctuations in performance between seasons. This pattern may reflect several underlying factors, including declining physical durability, injury risk, or changes in pitching mechanics as pitchers age.

In addition to model-based aging curves, my analysis also examines the raw distribution of peak performance ages as seen in *Figure 4*. Using the lowest ERA season for each pitcher with sufficient workload, the peak-age distribution shows that most pitchers achieve their best performance between ages 25 and 30.

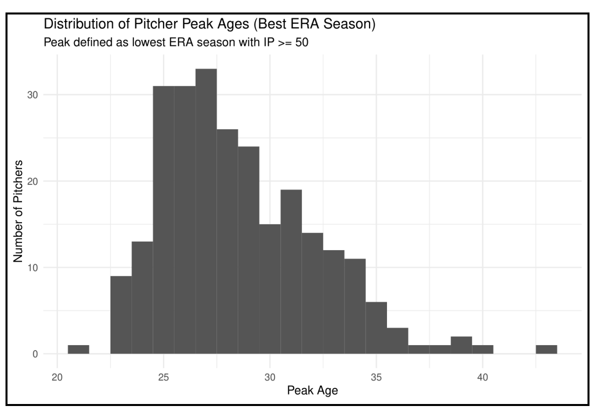

Figure 4: Distribution of Pitcher Peak Ages (Best ERA Season)

The histogram of peak ages shows a clear clustering around the late twenties, with the median peak occurring around age 28. Very few pitchers reach peak performance after age 35, reinforcing the idea that pitcher performance tends to plateau and decline after the early thirties.

To understand how modern pitcher development compares with historical patterns, similar mixed-effects models were estimated using historical pitching data from the Lahman database covering the pre-Statcast era. These models estimate aging curves using ERA while controlling for strikeout rate, walk rate, innings pitched, and seasonal effects.

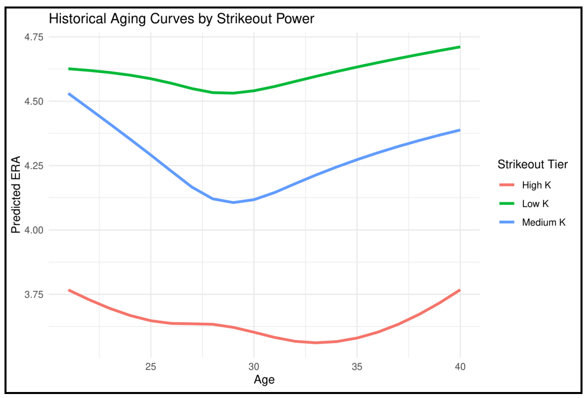

Figure 5: Historical Aging Curves by Strikeout Power

The historical aging curve above in *Figure 5* shows a slightly different performance trajectory compared to the modern Statcast-era curve. Pitchers in the historical sample appear to maintain slightly lower predicted ERA values later into their careers, suggesting that performance decline may have been more gradual in earlier baseball eras.

To further explore differences in pitching style historically, pitchers were grouped into strikeout tiers based on their strikeout rate. The resulting model estimates separate aging curves for low, medium, and high strikeout pitchers.

The results, as shown in *Figure 6,* show that high-strikeout pitchers consistently maintain lower predicted ERA values across the entire age range compared with medium and low strikeout pitchers. This suggests that pitchers who possess strong swing-and-miss ability have a sustained advantage in run prevention throughout their careers.

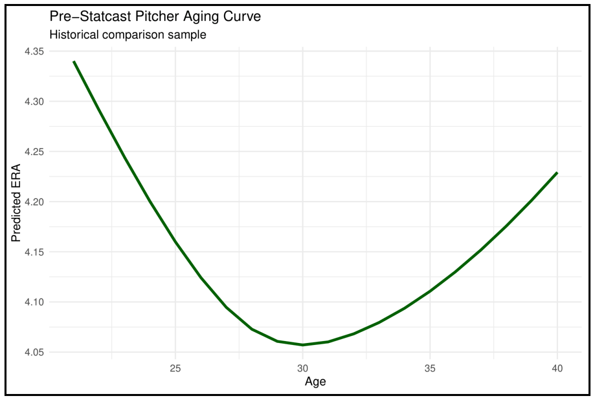

Figure 6: Pre-Statcase Pitcher Aging Curve

Finally, a combined model incorporating both historical and modern datasets allows direct comparison between the two eras. This model estimates separate aging curves for pitchers in the pre-Statcast and Statcast environments while controlling for strikeout rate, walk rate, innings pitched, and year effects, which can be visualized in *Figure 7.*

The comparison suggests that pitchers in the modern Statcast era may reach peak performance earlier than pitchers in the historical sample. Model estimates indicate that the historical peak age occurs around age 30, while the Statcast-era peak appears closer to age 24. Additionally, the estimated decline rate after age thirty is steeper in the Statcast era, suggesting that modern pitchers may experience faster performance deterioration later in their careers.

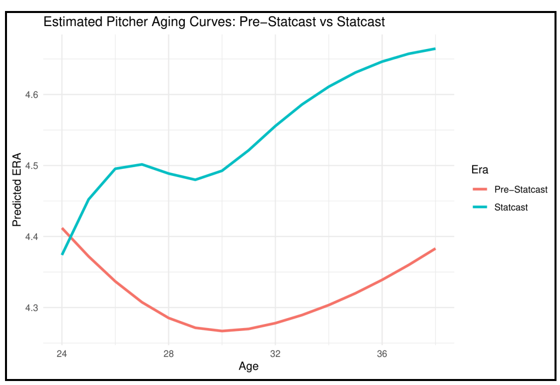

Figure 7: Estimate Pitcher Aging Curves: Pre-Statcase vs Statcast

Taken together, these results suggest that while pitcher aging patterns remain broadly consistent across eras, modern pitchers may reach peak effectiveness earlier and experience faster performance decline. The importance of velocity and strikeout ability across both historical and modern models also highlights the continued importance of dominance and pitch quality in sustaining long-term pitching success.

Conclusion

This study investigates how Major League Baseball pitcher performance evolves with age using modern Statcast-era data and historical pitching records. Using mixed-effects models, the analysis finds that pitcher performance generally improves early in a career, stabilizes during the late twenties, and gradually declines with age. The results also indicate that pitching characteristics such as velocity and strikeout ability play important roles in shaping aging patterns.

The comparison between historical and modern datasets suggests that pitcher aging may be occurring differently in the Statcast era. Modern pitchers appear to reach peak performance earlier and experience slightly faster performance decline after age thirty. Additionally, pitchers with higher velocity and stronger strikeout ability maintain lower predicted ERAs across the aging curve, highlighting the importance of pitch quality and dominance in sustaining long-term effectiveness.

Several limitations should be acknowledged. The modern dataset covers a relatively short time period compared to historical baseball data, and ERA may not capture all aspects of pitcher performance. Future research could incorporate additional Statcast metrics such as pitch movement, spin efficiency, or pitch usage patterns to further analyze how modern pitching development influences aging trajectories.

Overall, this study contributes to the literature by combining modern Statcast-era data with historical datasets and applying mixed-effects modeling to estimate pitcher aging curves while controlling for workload and pitching characteristics. These findings may help teams better understand pitcher development patterns and make more informed decisions regarding player contracts, workload management, and long-term roster planning.

References

Mercier, M.-A., Tremblay, M., Daneau, C., & Descarreaux, M. (2020). Individual factors associated with baseball pitching performance: scoping review. *BMJ Open Sport & Exercise Medicine*, *6*(1), e000704–e000704. https://doi.org/10.1136/bmjsem-2019-000704

Dun, S., Fleisig, G. S., Loftice, J., Kingsley, D., & Andrews, J. R. (2007). The relationship between age and baseball pitching kinematics in professional baseball pitchers. *Journal of Biomechanics*, *40*(2), 265–270. https://doi.org/10.1016/j.jbiomech.2006.01.008

APA PsycNet. (2026). Retrieved January 27, 2026, from Apa.org website: https://psycnet.apa.org/fulltext/1994-39406-001.html

Peak athletic performance and ageing: Evidence from baseball. (2026). *Journal of Sports Sciences*. https://doi.org/10.1080//02640410802691348

Coordination pattern of baseball pitching among young pitchers of various ages and velocity levels. (2016). *Journal of Sports Sciences*. https://doi.org/10.1080//02640414.2015.1133923